Bulletin of the Torrey Botanical Club 121(1), 1994, pp. 47–61

The role of disturbance, topography, and forest structure in the development of a montane forest landscape[1]

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ABSTRACT

HADLEY, K. S. (Department of Geography, Western Oregon State College, Monmouth, OR 97361). The role of disturbance, topography, and forest structure in the development of a montane forest landscape. Bull. Torrey Bot. Club 121: 47–61. 1994.—Human set fires beginning in the mid 1800s and repeated insect outbreaks of western spruce budworm (*Choristoneura occidentalis* Free.) and Douglas-fir bark beetle (*Dendroctonus pseudotsugae* Hopk.) during the past 50 years have resulted in a dramatic change in the montane (<ca. 2900 m) forest landscape of the Colorado Front Range. Here, I examine the historical and spatial relationship between these disturbance agents and topography using stand structure and dendroecological data from 38 contiguous stands. These data suggest that aspect and relief are important factors determining the spatial and temporal patterns of disturbance, succession, and rates of stand development. The rate of postfire stand development and hence, subsequent stand susceptibility to insect outbreaks appears to be related to aspect. North-facing stands experience rapid postfire development and greater susceptibility to insect attack due to higher host tree densities, larger mean tree size, and a more uniform distribution of host trees over larger contiguous areas. Postfire stand recovery on south facing slopes appears to be slower and stand susceptibility to insect attack is less due to lower host densities, smaller mean tree size, and a less uniform distribution of host trees over smaller areas. Relief, independent of aspect, enhances the structural diversity of the forest landscape by promoting irregular burn patterns and intensities, thus creating a fire-induced mosaic of different aged stands. As these different aged stands continue to grow older, they reach a stage of development susceptible to insect outbreaks at different times. As a result, insect-induced changes in the structural characteristics of the current landscape emulate fire-induced landscape patterns that developed largely due to human activities beginning in the 1860s.

Key words: disturbance, montane forest, landscape, stand structure, succession, Colorado Front Range, dendroecology.

The spread of disturbance across a forested landscape is largely determined by ecological processes and the physical characteristics of the landscape. Ecological processes, such as succession and competition, may cause forest stands or communities to be more susceptible to disturbance by creating a more vulnerable structure over time (e.g., Romme 1982; Hadley and Veblen 1993). Topography also influences disturbance by influencing environmental gradients and by increasing site susceptibility to certain types of disturbance, such as lightning strikes and windthrow (Foster 1988). Whereas ecological processes may determine the timing of disturbance across a landscape, topography will likely determine its spatial pattern (Swanson et al. 1988).

Historically, ecologists have focused on the influence of processes on the development of pattern across a landscape (e.g., Cowles 1899; Watt 1947). More recently, proponents of landscape ecology have sought to understand how pattern influences process (Turner 1989). Whereas it is important to consider the influence of ecological processes on landscape development, it is equally important to consider the role of topography in governing ecological processes, including disturbance. For example, Foster and Boose (1992) have shown that the rate of spread and different intensities of disturbance across a landscape are

[1] Support was provided in part by grants from the Colorado Mountain Club, the Graduate School at the University of Colorado, Grant No. SES 8618621 to T. T. Veblen from the National Science Foundation, and a faculty development award from Western Oregon State College.

[2] I thank Dr. T. T. Veblen for his guidance and support, and the National Park Service for permission to conduct this research and for providing me access to Park Service records. Research assistance was provided by A. Steele, J. Morenz, D. Goldblum, and M. S. Reid. I am grateful to D. Lorenz for creating the maps for this project and to M. Savage, D. K. Yamaguchi and two anonymous reviewers who provided helpful comments on an earlier version of this manuscript.

Received for publication June 30, 1993, and in revised form September 11, 1993.

controlled by the pre-disturbance structure of a landscape and its relationship with the underlying topography.

Synergism among disturbance agents must also be considered when examining the spatial and temporal patterns of disturbance across a landscape. Although disturbance agents may appear to act independently they are often causally related (Knight 1987). Drought, for example, increases the probability of both fire (Schroeder and Buck 1970; Brotak 1983) and insect outbreaks (Mattson and Haack 1987). Similarly, insect-induced tree mortality, especially in forested regions with continental climates, increases fuel availability and the probability of severe fires (Dodge 1972). Hence, meaningful studies of disturbance involving more than one disturbance agent must not only consider the effects of each agent but also their combined effects and how these agents interact with the physical environment.

The montane forests of the Colorado Front Range provide an opportunity to examine the interaction of disturbance, stand development and topography as they act to shape the contemporary forest landscape. During the past one hundred and fifty years, human set fires and outbreaks of western spruce budworm (*Choristoneura occidentalis* Free.) and Douglas-fir bark beetle (*Dendroctonus pseudotsugae* Hopk.) have acted alone and in combination to alter the character of the montane (i.e., <2900 m) landscape of the Colorado Front Range.

In this study, I examine the fire and insect outbreak histories for 38 contiguous stands in the Cow Creek Drainage of Rocky Mountain National Park. My objective is to describe the temporal and spatial relationships between fire and insect outbreaks as they pertain to local variations in topography and forest structure. Specifically, I hypothesize that the current forest structure and the spatial characteristics of recent insect outbreaks are the result of topographically controlled burn patterns that developed following the arrival of European settlers during the late 1800s.

SUCCESSION AND DISTURBANCE IN THE FRONT RANGE MONTANE FORESTS. The montane forests of the Colorado Front Range are part of a landscape mosaic shaped by steep environmental gradients and frequent natural and anthropogenic disturbance (Peet 1978, 1981, 1988; Veblen and Lorenz 1986, 1991). In general, succession patterns in these forests vary according to topographic position, edaphic conditions, and propagule availability following disturbance (Peet 1981). Post-disturbance succession primarily follows an "initial floristics" pattern (sensu Egler 1954), where late-successional species establish synchronously with early successional species (Veblen and Lorenz 1986).

FIRE. Disturbance studies in these forests are few and have focused primarily on fire (e.g., Clements 1910; Clagg 1975; Rowdabaugh 1978; Laven et al. 1980; Skinner and Laven 1983; Veblen and Lorenz 1986; Goldblum and Veblen 1992). These studies generally identify three historical fire eras based on fire frequency, including: 1) a pre-settlement era (pre-1860), 2) a settlement era beginning ca. 1860 (Mills 1911), and 3) a fire suppression era that began in the late 1800s (Anonymous 1988) but was not officially implemented by the National Park Service until the 1920s (Ise 1961). The settlement era is distinguished from the earlier and late periods primarily by its higher fire frequency and its greater burned area. As a consequence of the widespread burning during the settlement era, much of the present montane forest in the Front Range consists of young (< 130 yr) postfire stands of similar structure and composition (Veblen and Lorenz 1986).

The implementation of fire suppression policies beginning around the turn of the century also increased the homogeneity of these forests by increasing the dominance of shade tolerant species such as Douglas-fir (*Pseudotsuga menziesii*). During earlier periods of higher fire frequency, seedlings and saplings of these species were eliminated by ground fires.

WESTERN SPRUCE BUDWORM AND DOUGLAS-FIR BARK BEETLE. Compared to fire, insect outbreaks are less well studied in these forests, yet their effects on forest structure are in many ways comparable (Hadley and Veblen 1993). The two most destructive insects in these forest are western spruce budworm and Douglas-fir bark beetle.

Western spruce budworm is a defoliating insect that causes growth suppression and tree mortality by preventing photosynthesis over prolonged periods. Small suppressed trees, saplings, and seedlings are most susceptible to budworm-induced injury and mortality due to their limited energy reserves. In addition, these subcanopy trees are often subjected to higher concentrations of budworm larvae that disperse downward through the canopy to understory hosts (Carlson et al. 1983; Fellin et al. 1983).

Western spruce budworm also feeds on the cones and seeds of Douglas-fir (Dewey 1970; Fellin et al. 1983). During outbreaks, western spruce budworm can eliminate 100% of the available seed source and retard seedling recruitment for several years following outbreaks (Fellin et al. 1983). Budworm attack also causes fine-root mortality (Swaine and Craighead 1924; Redmond 1959), and increases tree vulnerability to subsequent pathogen (Stillwell 1956) and bark beetle attack (Carlson et al. 1983).

Douglas-fir bark beetle is a cambial feeder that typically attacks weakened, large-diameter trees with thick phloem suitable to larval development (Cates and Alexander 1982). Tree mortality is the result of girdling and the introduction of woodstaining fungi that inhibit water and nutrient transfer to the upper portions of the tree.

The development of spruce budworm and Douglas-fir bark beetle outbreaks in the Front Range appear to be dependent on stand vulnerability as determined by host tree density and forest structure (Hadley and Veblen 1993). High host density is important because it increases the probability of successful insect dispersal between host trees and it ensures an abundant food source for these insects. Dense, postfire, multistoried stands of Douglas-fir are especially susceptible to budworm attack whereas older, postfire, even-aged stands are susceptible to sever beetle outbreaks (Berryman 1982).

The onset and severity of insect outbreaks are also related to environmental factors. Drought is probably the most important abiotic factor predisposing susceptible stands to outbreaks due to its regional extent and influence on a multitude of other disturbance factors (Mattson and Haack 1987). Given similar stand structures, budworm- or beetle-induced tree mortality is most severe in habitats that exacerbate drought conditions, such as low elevation, south-facing slopes with shallow, coarse-textured soils (Fellin et al. 1983; Christiansen et al. 1987).

Methods. SITE DESCRIPTION. The study area (ca. 10.5 km^2) is located along the east-west axis of the non-glaciated middle basin of the Cow Creek drainage between 2450 m and 2750 m elevation (Fig. 1). Slope angles are generally steep (mean 20°, SD 6.6°, range 3–31°) and the most common aspects are north to northwest (42%) or south to southeast (29%). The southern and northern slopes of the basin are underlain by Silver Plume Granite which is expressed as tor-like topography in the southeast quadrant of the study area.

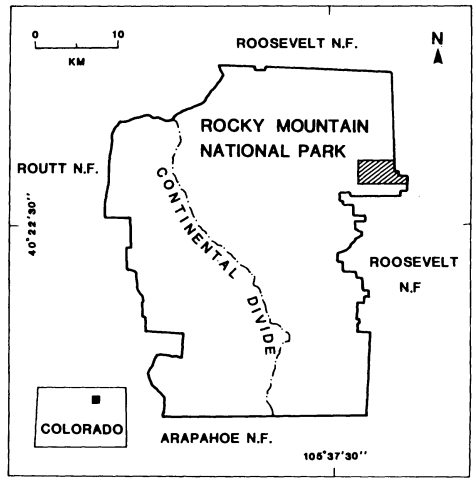

Fig. 1. Study area (lined) in the northeastern portion of Rocky Mountain National Park, Colorado.

Soils within the study area are broadly classified as Cryoboralfs, Cryocrepts, and Cryorthents. The relatively weak soil development at many sites probably reflects the combined effects of steep slopes and repeated erosional episodes following earlier fires (Morris and Moses 1987). The mean annual precipitation for the study area is approximately 41 cm (National Park Service records 1984). The seasonal distribution of precipitation shows a pronounced May maximum due to strong convective activity and a winter minimum (Barry 1973).

Park Service records (Anonymous 1936–1986) document epidemics of western spruce budworm and Douglas-fir bark beetle in or near Rocky Mountain National Park since the late 1930s. Western spruce budworm outbreaks occurred during the 1938–1945, 1958–1965, and 1976–1985 periods. Douglas-fir bark beetle outbreaks occurred during the 1950–1951 period, and again from 1984 to present.

The fire history of the montane forests in Rocky Mountain National Park is generally known from historical records and studies using tree ring data (Clements 1910; Clagg 1975; Rowdabaugh 1978; Skinner and Laven 1983). Although these studies are based on small sample sizes (i.e., fewer than 25 scarred trees), their results clearly identify the pre-settlement, settlement, and fire suppression eras noted above. The mean fire return interval for the pre-settlement period for these forests is estimated to be 40 years (Rowdabaugh 1978).

Field Methods. Thirty-eight mixed stands, consisting largely of Douglas-fir, ponderosa pine (*Pinus ponderosa*), lodgepole pine (*Pinus contorta*) and aspen (*Populus tremuloides*) were studied between 1988–1990 (Fig. 2). Stand boundaries were initially interpreted from 1:15,840 color aerial photographs using stand geometry, size structure, canopy density, and species composition as mapping criteria. Field observations of tree mortality resulting from the most recent outbreaks of western spruce budworm (1974–1985) and Douglas-fir bark beetle (1984–present), aided in the further delineation of stands having similar size structures but different species compositions. The study area was also divided into a southern (CCS; stands S-1 to S-21) unit and a northern (CCN; stands, N-1 to N-17) unit to facilitate comparisons between stands occupying north facing versus south facing aspects. The boundary between the units is the riparian vegetation associated with the main channel and upper southern tributary of Cow Creek (Fig. 2).

Sampling employed 10 × 10 to 10 × 60 m plots subjectively located in each of the mapped stands. The number and size of plots for each stand varied according to stand size and composition, tree size, tree density, and the spatial distribution of trees within the stand. Large stands or stands with highly clumped tree distributions were sampled with multiple plots; open stands with regularly spaced trees and dense stands were sampled with a single large or a single small plot respectively. The number of trees per plot varied from 12 to 72 (mean = 26).

Data collected in each plot included diameter at breast height (DBH) measurements of all live and dead-standing trees >4 cm DBH, and counts of live and dead seedlings (<1.4 m tall) and saplings (>1.4 m tall but <4 cm DBH). All live trees in each plot were cored at ca. 30 cm height. Age determinations for seedlings were not made because of the destructive methods required to obtain these data; stand ages therefore represent minimum ages and are estimated to be 15–20 years less than the true stand ages. Supplemental cores were collected from live trees within 20 m of those plots having high host mortalities or low densities of codominant tree species for additional age determinations.

Non-forested areas and areas of open ponderosa pine woodland were inspected for the presence of logs and stumps, seedlings and saplings, and site characteristics. Tree cores from five or more of the largest ponderosa pine in these areas were collected to obtain stand ages.

Spearman rank correlations were calculated for the northern and southern sample units to determine the strength of association between the number of dead host trees, total host tree density, total basal area, the number of dead host seedlings and the number of dead host saplings. A rank sum two sample test (Mann-Whitney) was also calculated to determine if the total number of host trees and the number of dead trees sampled in the southern (CCS) and northern (CCN) units represent the same population.

Fire dates for stands with fire scarred trees were determined using tree cores (Barrett and Arno 1988; Goldblum and Veblen 1992) extracted from fire scarred trees identified by systematically traversing each stand. Several of the soundest single scarred trees and all of the multiple scarred trees encountered in each stand were sampled. Because earlier fire dates derived from multiple scarred trees using tree cores are unreliable (Barrett and Arno 1988), fire dates are only assigned to the most recent scars. Fire dates determined using increment borer methods are accurate within 2 years of the actual fire date (Goldblum and Veblen 1992). Park Service policy precluded the use of cross-sections or wedges for the dating of fire scars.

Dendroecological Analyses. Age structures, growth suppressions and releases, and ring-width chronologies were developed from a total of 684 tree cores (mean number of trees per stand = 18). All cores were mounted and sanded with progressively finer sandpaper (Stokes and Smiley

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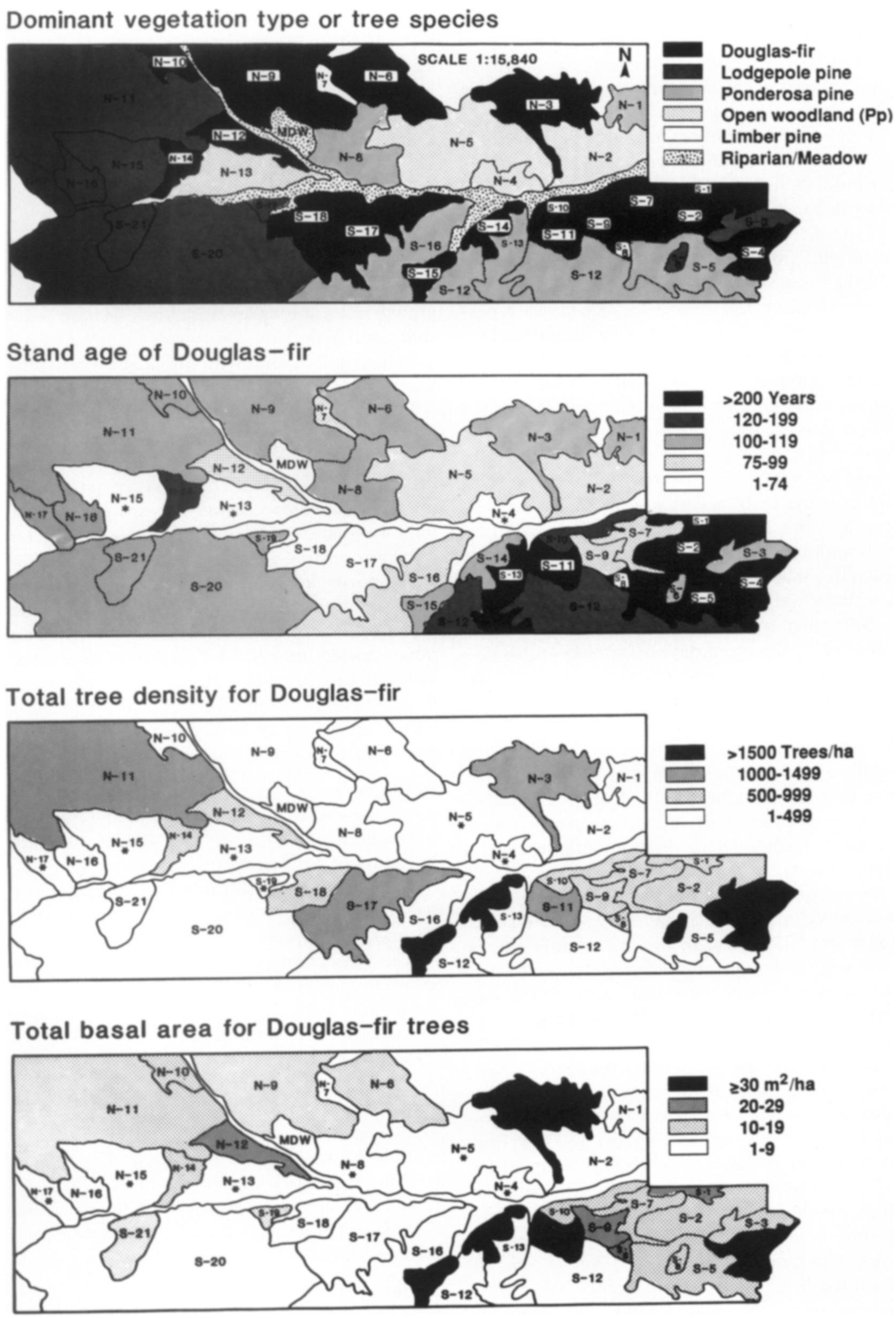
Figs. 2–5. 2. Vegetation types based on the canopy dominant tree species as determined by basal area and(or) density. The boundary between the southern (CCS) and northern (CCN) study units is the riparian vegetation associated with the main channel and upper southern tributary of Cow Creek. — 3. Minimum stand ages for Douglas-fir (Tables 2a and 2b). Asterisks indicate Douglas-fir are absent or rare. — 4. Total tree densities (live plus dead) for Douglas-fir. Asterisks indicate values <1. — 5. Total basal areas (live plus dead) for Douglas-fir trees. Asterisks indicate values <1.



Dominant vegetation type or tree species
SCALE 1:15,840
N
Douglas-fir
Lodgepole pine
Ponderosa pine
Open woodland (Pp)
Limber pine
Riparian/Meadow
Stand age of Douglas-fir
>200 Years
120-199
100-119
75-99
1-74
Total tree density for Douglas-fir
>1500 Trees/ha
1000-1499
500-999
1-499
Total basal area for Douglas-fir trees
≥30 m²/ha
20-29
10-19
1-9

1968) before ring counting and measuring under stereomicroscopes.

Years corresponding to the onset of growth suppression or release were noted for each core during counting. A growth suppression is defined as an abrupt reduction in mean ring width of >40% when sustained over a period of 5 years or more (Schweingruber 1986). A growth release was defined as an abrupt increase in mean ring width of >250% when compared with the mean ring width of the prior 5 years (Veblen et al. 1991).

Suppression and release data are given as percentages of live trees that were suppressed or released following earlier periods of canopy disturbance and that were alive at the time of sampling. Because defoliation often results in missing rings, exact dates of the onset of defoliation for each tree and the number of years with missing rings are not known. Consequently, these data underestimate the severity of the budworm outbreaks. The extreme difficulty of cross-dating subcanopy, budworm-defoliated, Douglas-fir has been documented elsewhere (Swetnam 1987) and precluded the use of this technique here.

Suppression frequencies provide a qualitative measure of the onset date of defoliation, its duration, and the relative severity of defoliation. Increased suppression frequencies within a given time period imply increased outbreak severity. Release frequencies provide a similar measure of host recovery following canopy disturbance due to defoliation, beetle attack or fire; they also provide a measure of the growth responses of nonhost species to defoliation and insect-induced mortality of competing host trees.

Mean ring-width chronologies for Douglas-fir were developed from 5 randomly selected cores for each of 33 stands. In two stands only 4 trees were used to develop chronologies (N-2 and S-19) and no chronologies were developed for 3 stands (N-3, N-8, and S-19) due to the local absence or scarcity of Douglas-fir.

Ring widths were measured to the nearest 0.001 mm using a Henson measuring machine. The resulting series were standardized using horizontal lines and averaged using the INDEX and SUMAC computer programs (Fritts 1976; Graybill 1982). Standardization involves fitting a line to an observed ring-width series and computing a dimensionless index value. Index values are calculated by dividing observed ring widths by the fitted line. Following experimentation with several standardization options, (e.g., polynomial, exponential, and cubic spline function), the horizontal-line standardization was chosen because it scales the variances both within and among the cores but does not remove all low-frequency variations in the time series. This facilitates the detection of deviations from average growth rates, such as those associated with major canopy disturbances including defoliation or tree death (Veblen et al. 1991).

Each of the 33 ring-width chronologies was grouped into one of seven chronology types where each chronology type represents a similar radial growth trend among several stands. Marker years representing periods of severe drought were used to help identify stands with similar growth trends.

Results. The spatial arrangement of forest, woodland, riparian, and meadow vegetation types suggest that aspect, elevation and topography are primary determinants of the local vegetation mosaic (Fig. 2). Ponderosa pine and open ponderosa pine woodlands occupy the most xeric sites on the lower portions of south facing slopes and on steep, north facing slopes having shallow soils and large areas of exposed granite bedrock. The more mesic lower portions of the north facing slopes are dominated by Douglas-fir as are the higher elevation stands on the south facing slopes. At the highest elevations located in the western portion of the study area, lodgepole pine is the dominant tree species.

FIRE HISTORY. Fire scar dates and age structure data indicate that each of the sampled stands experienced one or more fires during the past 300 years. A total of 27 fire scar dates were obtained from 16 stands (Table 1). The number of fires for the pre-settlement era (4), settlement era (2), and fire suppression era (5) does not suggest a clear distinction between these periods based on fire frequency alone. However, stand ages indicate that a fire in 1863 altered most of the stands within the study area.

Mean fire intervals (MFI) for the pre-settlement, settlement, and fire suppression eras in the study area are 29 (median 29; range 20–38) years, 20.7 (median 23; range 13–26) years, and 15.3 (median 17; range 8–21) years respectively. The MFI for the entire chronology (1747–1971) is 22.4 years (median 22 years). During the 1938–1985 period of insect outbreaks, three small fires occurred in the study area.

FOREST STRUCTURE. Minimum stand ages for the study area range from 57 to 266 (Tables 2a

Table 1. Fire scar dates for the Cow Canyon study area. The number of scars represents the total number of scars sampled in the study area for a given fire year. The fire interval is the number of years between successive fires.

	Fire year										
	Pre-settlement era				Settlement era		Fire suppression era				
	1747	1767	1796	1834	1863	1886	1912	1925	1946	1963	1971
Stand	N-3	S-12	S-12	S-11	N-3	S-14	N-1	N-1	N-9	N-7	S-13
		S-13	S-14	S-13	N-6		N-17	N-14	S-2		
					S-5		S-13	S-12	S-10		
					S-16		S-20				
# scars	1	2	2	2	4	1	5	4	3	1	2
Interval		20	29	38	29	23	26	13	21	17	8

and 2b). Stands with minimum ages of >125 years are limited to the southeastern quadrant of the study area. Older (>200 yr) ponderosa pine are also present in several stands throughout the study area (Fig. 2; Tables 2a and 2b). Fire scar data (Table 1) and the young ages of stands comprising the northern unit and southwestern one-half of the southern unit (Tables 2a and 2b; Fig. 3) indicate that much of the study area burned during the fire of 1863.

Generally, the ages of nonhost tree species within a given stand are comparable to those of Douglas-fir (Tables 2a and 2b). This is especially true among the younger stands (<150 years old). There are, however, several exceptions where remnant limber pine (*Pinus flexilis*; S-1) and ponderosa pine (S-12, S-13, N-3, N-6, N-9, N-14) exceed the oldest Douglas-fir by >100 years. Similarly, the oldest Douglas-fir in stand S-5 is >150 years older than the oldest ponderosa pine (Table 2a). The occurrence of these older individuals indicates the presence of several postfire cohorts in the study area.

The total densities (live and dead trees) and basal areas of Douglas-fir are highest among young, postfire stands in the southern sample unit (Figs. 4 and 5). High total basal area values were noted among stands with various size structures: young, dense stands (e.g., S-14 and S-15); young, less dense, nearly pure stands of Douglas-fir (N-12); old, less dense stands with larger stem sizes (S-4); and, stands with a wide range of stem sizes (N-3). Live and dead-standing tree, sapling, and seedling densities and basal areas for each tree species in each stand are reported elsewhere (Hadley 1990).

Spearman rank correlations show that the number of dead trees per stand is highly correlated to total host tree density and basal area for both the southern unit (r_s = 0.765 and 0.641; P < 0.005) and the northern unit (r_s = 0.643 and 0.628; P < 0.01; Tables 3 and 4). However, statistical comparisons (Mann-Whitney *U*-Test) of the total number of trees and the number of dead trees to aspect, indicate that these two units represent different populations (P = 0.006; Table 5).

The relative densities of insect-induced tree, seedling, and sapling mortality (i.e., percentages of trees, seedlings, and saplings that are dead) of Douglas-fir following the most recent sequence of insect outbreaks, vary considerably among the sampled stands (Figs. 6 and 7). Tree mortality is highest (81–100%) in stands S-1, S-5, S-6, S-10, S-14, S-15, all of which occupy north facing sites and have high host densities and basal areas (Fig. 6). No mortality is present in four stands (N-1, N-2, N-7, N-16) characterized by southerly aspects, low host densities, and low host basal areas. Similarly, three stands with northerly aspects, S-18, a young stand (47 yrs), and S-16 and S-21 with low host densities and basal areas, have no host tree mortality (Fig. 6). Additional stands with low host tree mortality (<20%) include stands that are young (S-17), have low host densities (N-6, N-10), or occupy mesic sites (S-9, N-10).

Budworm-induced seedling and sapling mortality is highest in young stands having high Douglas-fir tree densities, basal areas, and mortality (S-14, S-15, and N-3) or dense "doghair" stands (S-6; Figs. 4–7). Spearman rank correlations for the southern and northern sample units show the number of dead seedlings and the number of dead trees per stand are moderately to strongly correlated (CCS: r_s = 0.523; P < 0.01; CCN: r_s = 0.637; P < 0.005; Tables 3 and 4). In both sample units, the number of dead saplings is more highly correlated to total host tree density and basal area than is the number of dead

Table 2a. Minimum stand age (yr) for live Douglas-fir trees in southern study unit (CCS). The number of trees aged is in parentheses after the age range for the respective species. Ages are not given for non-host species where n is <5 except for remnant trees (asterisks). Species abbreviations are PM, *Pseudotsuga menziesii*; PC, *Pinus contorta*; PP, *Pinus ponderosa*; PF, *Pinus flexilis*; PT, *Populus tremuloides*. Mean minimum stand age, southern study unit, is 145; SD = 65; range = 57–266.

Stand	S-1	S-2	S-3	S-4	S-5	S-6
Stand age	240	266	112	254	201	113
PM	138–240 (9)	26–266 (11)	76–112 (23)	172–254 (15)	182–201 (6)	90–113 (7)
PM*	—	—	—	—	347 (1)	—
PP	101–234 (7)	—	—	—	104–214 (13)	—
PF*	387–392 (2)	273 (1)	—	—	—	—
PC	—	—	103–108 (28)	—	—	90–115 (7)
Stand	**S-7**	**S-8**	**S-9**	**S-10**	**S-11**	**S-12**
Stand age	93	221	88	161	205	123
PM	53–93 (9)	21–221 (5)	26–88 (15)	110–161 (7)	71–205 (5)	25–123 (5)
PM*	181–203 (2)	—	161–200 (2)	214 (1)	304–361 (3)	161–201 (3)
PP	—	137–214 (8)	—	103–200 (5)	—	117–402 (11)
PP*	—	—	—	—	321 (1)	—
PT	—	—	99–152 (7)	—	—	—
Stand	**S-13**	**S-14**	**S-15**	**S-16**	**S-17**	**S-18**
Stand age	204	111	104	96	61	57
PM	129–204 (8)	89–111 (6)	90–104 (7)	29–96 (6)	30–61 (19)	39–57 (17)
PM*	—	234–275 (2)	—	—	—	—
PP	109–230 (6)	100–108 (5)	—	142–182 (6)	—	—
PP*	311 (1)	249–305 (3)	—	—	—	—
PF	—	—	—	—	—	63–86 (5)
Stand	**S-19**	**S-20**	**S-21**			
Stand age	107	115	103			
PM	87–107 (4)	67–115 (13)	77–103 (8)			
PC	90–109 (5)	53–119 (14)	63–109 (10)			

host seedlings (Tables 3 and 4). The relationship between the number of dead saplings and the number of dead trees is only significant in the southern sample unit ($r_s = 0.706$; $P < 0.005$; Tables 3 and 4).

TREE-GROWTH RESPONSES. Suppression and release frequencies were calculated for the southern and northern study units to ensure adequate sample sizes and to allow comparisons between their contrasting environmental conditions and stand structures.

The frequency and timing of Douglas-fir growth suppressions suggest that the 1976–1985 budworm outbreak was the most severe during the period of documented record (Fig. 8). Earlier periods of growth suppression are apparent in the southern unit (CCS) during 1943–1956 and in the northern unit (CCN) during 1958–1963. Release frequencies for Douglas-fir indicate accelerated growth following each of these earlier suppression periods (Fig. 8). The durations of suppression and release episodes are longer in the southern unit in all cases.

Release frequencies for ponderosa pine and lodgepole pine, nonhost tree species whose growth is most responsive to changes in canopy structure (Hadley 1990), reflect increased growth rates during the 1940s and the 1980s (Fig. 8). Ponderosa pine in the southern unit exhibits releases during both periods while releases of lodgepole pine are limited to the earlier period. Ponderosa pine in the northern unit, including those growing in open woodlands, show few periods of increased growth during the last 110 years. Release frequencies of lodgepole pine in the northern unit correspond to the 1976-present period of overlapping budworm and Douglas-fir bark beetle outbreaks.

Ring-width chronologies for Douglas-fir indicate a wide range of growth conditions in the southern and northern sample units (Fig. 9). Chronology types I and II show the least response to the recent insect outbreaks and reflect open growth conditions on south facing slopes (Type I) or growth in young, dense, single-storied stands codominated by lodgepole pine (Type II). The Type I (represented by stand N-1) chronology is characterized by a slower decrease in overall growth and greater annual growth fluctuations

Table 2b. Minimum stand ages (yr) for live Douglas-fir trees in the northern study unit (CCN). The number of trees aged is in parentheses after the age range for the respective species. Ages are not given for non-host species where n is <5 except for remnant trees (asterisks). Species abbreviations follow Table 2a. Mean minimum stand age, northern study unit is 104; SD = 9; range = 90–121.

Stand	N-1	N-2	N-3	N-4
Stand age	114	94	105	—
PM	54–114 (7)	60–94 (4)	57–105 (8)	—
PM*	—	208 (1)	—	—
PP	77–118 (16)	111–132 (4)	105–174 (4)	95–175 (5)
PP*	—	192 (1)	326 (1)	—

Stand	N-5	N-6	N-7	N-8
Stand age	90	105	96	—
PM	55–90 (6)	75–105 (8)	35–96 (6)	—
PP	76–101 (8)	89–339 (5)	85–95 (5)	87–115 (10)
PC	—	—	69–99 (6)	—
PF	—	—	86–93 (5)	—

Stand	N-9	N-10	N-11	N-12
Stand Age	101	101	113	98
PM	63–101 (10)	81–101 (7)	95–113 (8)	55–98 (8)
PP	78–114 (6)	51–107 (5)	—	—
PP*	236 (1)	—	—	—
PC	—	68–107 (5)	103–108 (8)	69–111 (8)

Stand	N-13	N-14	N-15	N-16	N-17
Stand Age	—	121	—	112	104
PM	—	81–121 (8)	—	86–112 (8)	91–104 (6)
PP	117–157 (5)	92–119 (3)	—	—	—
PP*	—	181–250 (2)	—	—	—
PC	—	101–112 (5)	94–117 (8)	109–119 (8)	99–116 (7)

than stands having closed canopies (Fig. 9). Chronology Type II (represented by stand S-6) is typical of a dense stand exhibiting rapid post-fire initial growth followed by very low growth rates upon canopy closure.

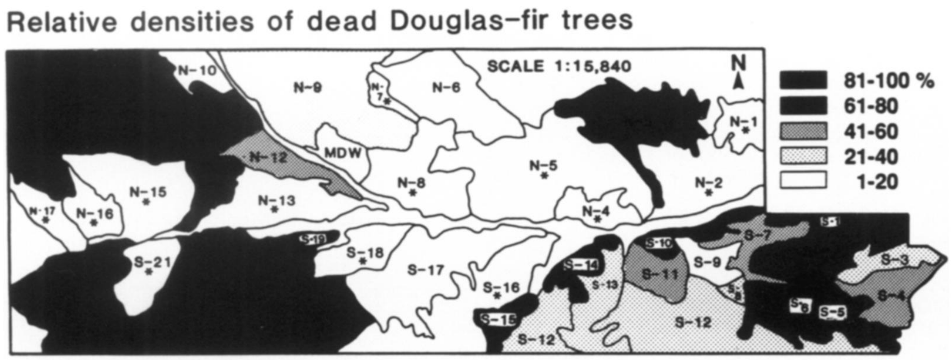
The Type III chronology (represented by stand S-2) exhibits a dramatic release during the late 1940s (Fig. 9). Windthrown and wind-snapped trees, large numbers of aligned logs, and growth releases among host trees indicate this release is a response to a large blowdown. Park Service records in 1949 (Anonymous 1936–86; Glidden 1982) and dendroecological data (Veblen et al. 1989) from the adjacent subalpine forest indicate blowdowns were common in 1947–1949. During the 1976–1985 budworm outbreak, stands represented by the Type III chronology experienced a sharp decrease in growth rates due to defoliation (Fig. 9). Post-outbreak growth rates have increased dramatically due to the low density of live canopy trees in these stands following blowdown, bark beetle mortality, or both (Fig. 6).

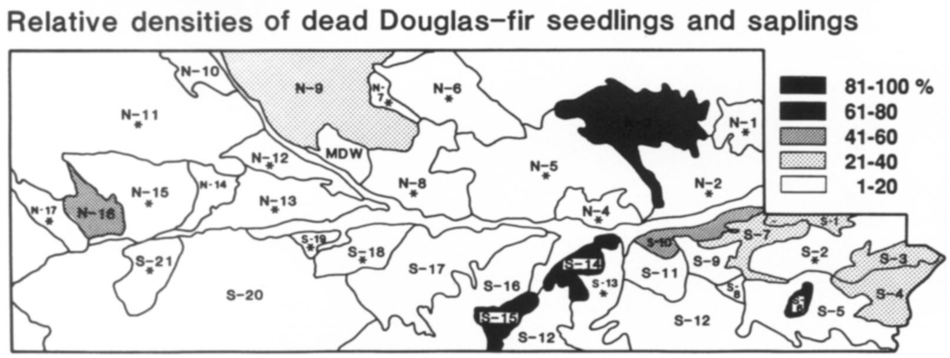
Chronology types IV, V, and VI display similar growth trends among different forest age structures (Fig. 9; Tables 6, 2a, and 2b). The Type IV chronology (represented by stand N-11) shows a sharp decrease in growth rates during the late 1930s and again in the late 1960s. The Type V chronology (represented by stand S-5) represents the oldest of the sampled stands. This chronology type exhibits slow initial growth before the 1863 fire followed by a subsequent increase in growth.

Table 3. Spearman rank correlation coefficients (corrected for ties) for stand variables; CCS sample unit. Number of stands = 21.

	Dead trees	Host tree density	Basal area	Dead seedlings	Dead saplings
Dead trees	1.000				
Host tree density	0.765***	1.000			
Basal area	0.641***	0.534**	1.000		
Dead seedlings	0.523**	0.445	0.227	1.000	
Dead saplings	0.706***	0.620***	0.510*	0.668***	1.000

* $P < 0.025$; ** $P < 0.01$; *** $P < 0.005$.





Figs. 6, 7. 6. Relative densities of dead Douglas-fir trees (percentage of dead trees). Asterisks indicate values <1. — 7. Relative densities of dead Douglas-fir seedlings and saplings (percentages of dead seedlings and saplings). Asterisks indicate values <1.

Growth rates decreased again during the 1950s and 1970s (Fig. 9). The Type VI chronology (represented by stand S-14) shows a rapid growth decline since ca. 1900 and two additional periods of reduced growth. Reduced growth occurred during the 1950s and late 1970s. The Type VII chronology (represented by stand N-3) exhibits a marked growth release more typical of nonhost species following the latest budworm and bark beetle outbreaks (Hadley and Veblen 1993). Douglas-fir surviving in these stands appear to have benefitted from reduced canopy competition due to defoliation and mortality.

Discussion. The forest structures of 38 contiguous stands in the Cow Creek Drainage primarily reflect the environmental influence of aspect and topography on the development of young (<150 years) fire initiated stands. Similar aged stands on north facing slopes have higher tree densities and basal areas than their south facing counterparts (Figs. 3–5), indicating that they

Table 4. Spearman rank correlation coefficients (corrected for ties) for stand variables; CCN sample unit. Number of stands = 17.

	Dead trees	Host tree density	Basal area	Dead seedlings	Dead saplings
Dead trees	1.000				
Host tree density	0.643**	1.000			
Basal area	0.628**	0.946***	1.000		
Dead seedlings	0.637**	0.524*	0.610**	1.000	
Dead saplings	0.381	0.608**	0.577*	0.726***	1.000

* $P < 0.025$; ** $P < 0.01$; *** $P < 0.005$.

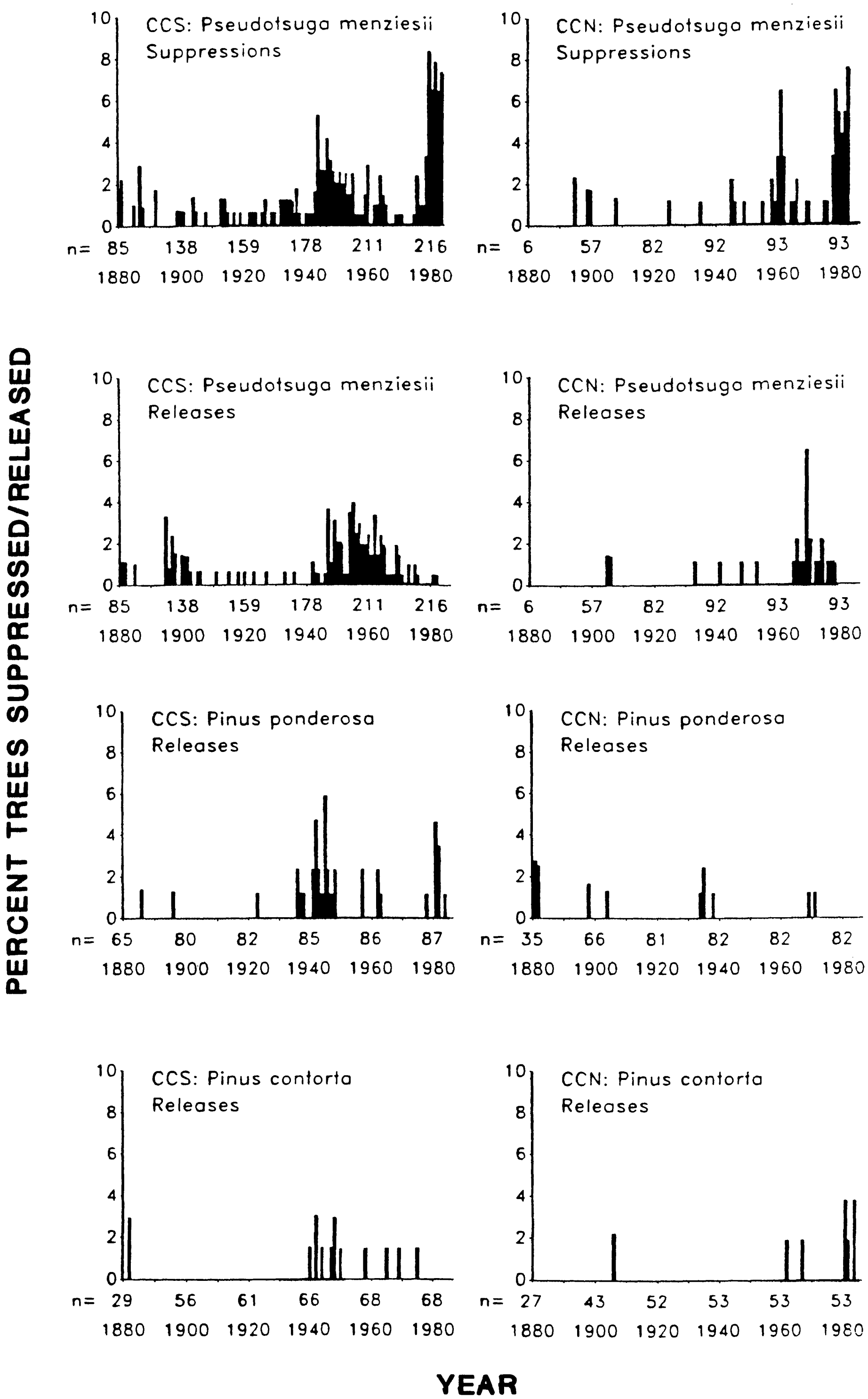


Fig. 8. Percentages of suppressed and released Douglas-fir and released ponderosa and lodgepole pine for the southern (CCS) and northern (CCN) sample units. See text for definition of suppression and release.

Table 5. Rank sum two sample test for host trees/stand and dead trees/stand by aspect (CCS versus CCN).

Aspect	Sample size	U-statistic	Average rank	U-statistic	Average rank
		Host trees/stand ($P = 0.006$)		Dead trees/stand ($P = 0.006$)	
CCS	21	272.5	24.0	271.5	23.9
CCN	17	84.5	14.0	85.5	14.0

reached a developmental stage susceptible to insect outbreaks more quickly. Stands occupying north facing slopes also have a greater range of stand ages (Table 2a), the result of a complex fire history in association with a topographically diverse terrain.

FIRE. The fire history of the Cow Creek Drainage is similar to that of much of the Front Range, indicating a period of frequent or severe fires during the period of European settlement beginning ca. 1860 (Skinner and Laven 1983; Veblen and Lorenz 1986; Goldblum and Veblen 1992). The regional importance of these fires was a large scale change in forest structure (Veblen and Lorenz 1991). At the landscape scale, fire served as a template for subsequent insect outbreaks by creating a mosaic of different aged stands. The mosaic is best developed in areas of greatest topographic variation due to uneven burn patterns that resulted in a higher proportion of remnant stands both in the Cow Creek Drainage and elsewhere in the Front Range (Goldblum and Veblen 1992).

Fire suppression has also contributed to the character of the montane forest landscape. However, the effect of fire suppression also appears to be related to aspect. On north facing slopes, fire suppression has promoted high densities of shade-tolerant Douglas-fir. On south facing slopes, the apparent effect of fire suppression is an increased rate of meadow invasion by Douglas-fir.

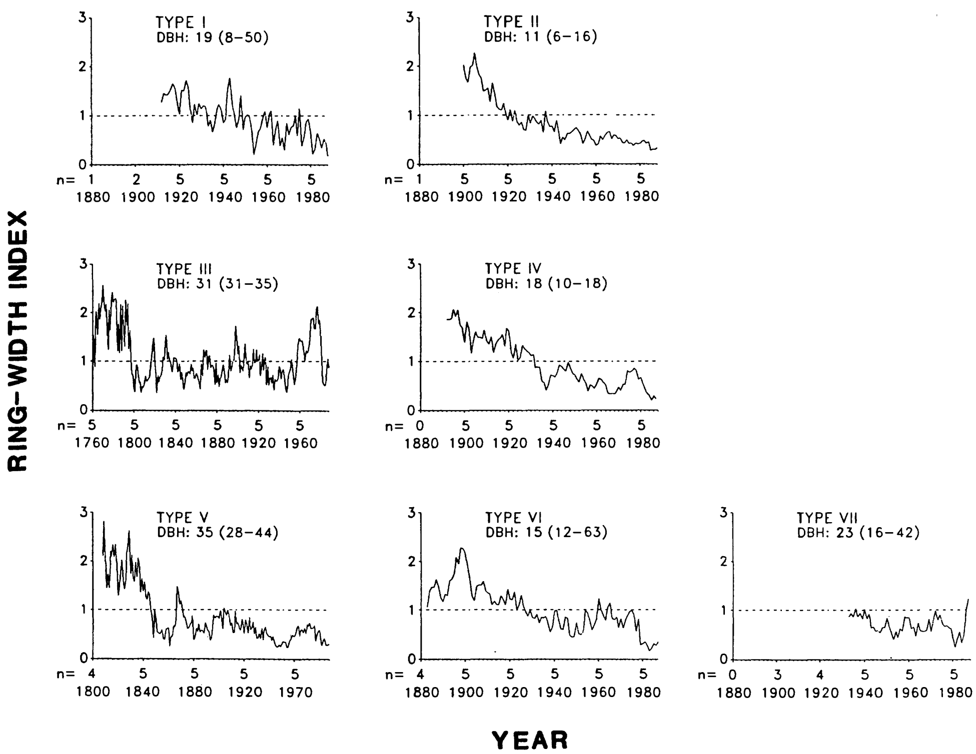


Fig. 9. Type Chronologies for randomly selected Douglas-fir trees. Each type represents one or more stands (see Table 4). The median and range (parentheses) of dbh values for the sampled trees are given. The mean index values for each chronology type are indicated by horizontal dashed lines. The minimum number of trees represented in a given decade is shown as the value of N below the X axis.

Table 6. Stands represented by Type Chronologies and their growth responses to environmental and stand conditions.

Type	Growth conditions	Stands
I	Open conditions on south-facing slopes	N-1, N-2, N-5, N-6, N-7, N-17, N-16
II	Dense, closed canopy conditions with small diameter trees	S-3, S-6
III	Blowdown-induced release in older closed canopy stands	S-1, S-2, S-7, S-8, S-11, S-12, S-13
IV	Young, mixed stands codominated by climax and seral species	S-19, S-20, S-21, N-10, N-11, N-12, N-14, N-16
V	Stands including old, fire-remnant trees with complex fire histories	S-4, S-5, S-9, S-16
VI	Young stands exhibiting high insect-induced mortality and defoliation	S-14, S-15, S-17
VII	Post insect-disturbance release due to canopy removal	N-3, N-9

Spruce Budworm Defoliation. Suppression and release frequencies (Fig. 8) and ring-width chronologies (Fig. 9) show that although severe budworm defoliation occurred on both north and south facing slopes, it was more common on north facing slopes. Increased defoliation and suppressed radial growth in these stands may be attributed to stand structure, specifically, higher host densities, greater variation in host size, and a more advanced stage of stand development. The continuous distribution of host trees among these stands also enhances the survivorship of both budworm and bark beetle during dispersal.

In Cow Canyon, the lower severity of budworm outbreaks in stands on south facing slopes (Fig. 8) may be attributed to several factors including lower host densities (Fig. 4) and the single-storied canopies of host tree populations. These stands are also generally younger than those experiencing high budworm and bark beetle-induced tree mortality on north facing slopes (Fig. 3; Table 2b). The presence of open woodland may also inhibit insect dispersal due to greater distances between host trees.

Bark Beetle-Induced Mortality. The spatial pattern of bark beetle-induced tree mortality in Cow Canyon is related to the fire history of the respective slopes and their different rates of re-establishment. Older, larger (hence more likely hosts), Douglas-fir trees on south facing slopes often occur in isolated stands and in lower densities than on north facing slopes (Fig. 4). Hence, bark beetle outbreak severity is generally greater on north facing slopes for similar reasons that budworm outbreaks are more severe on these slopes, i.e., higher host densities and a more even distribution of suitable (large diameter) trees. These results suggest that, during outbreaks, higher host tree densities on north facing slopes contribute to higher rates of successful insect attack than does the weakening of host trees on south facing slopes due to higher moisture stress (e.g., Cates and Alexander 1982).

Beetle mortality on north facing slopes appears to be associated with host age, density, basal area, and contagious stand conditions (Figs. 3–6). However, the relative importance of each of these characteristics is unclear. A partial explanation for this complex spatial relationship is the synergistic relationship between budworm and bark beetle disturbance. Stands that experience severe budworm damage (Fig. 8), later undergo a similar level of disturbance during subsequent bark beetle outbreaks (Fig. 6).

Topographic dissection in conjunction with fire has also contributed to the locally irregular pattern of bark beetle-induced mortality by promoting a mosaic of old-growth stands with lower densities of larger individuals (S-1, S-2, S-5; Fig. 4). Historically, these trees were important seed sources that accelerated postfire re-colonization of these sites and the recruitment of trees into insect-susceptible size classes.

Effects of Insect Outbreaks on Landscape Structure. The effects of budworm and bark beetle outbreaks at the landscape scale are similar to those occurring at the stand level (Hadley and Veblen 1993). There is a minor shift in dominance as a more homogeneous size and age structure develops among Douglas-fir populations. The net effect is an increase in the persistence of nonhost, shade-intolerant species through decreased competition. Concurrently, budworm and bark beetle outbreaks decrease host densities and create a narrower size frequency distribution of trees over large areas by eliminating large and small trees. These outbreaks also result in younger stand ages as older, large diameter, fire-remnant trees are eliminated by bark beetle attack.

In general, outbreaks of western spruce budworm and Douglas-fir bark beetle decrease the structural diversity of Douglas-fir populations across the landscape by eliminating both the younger, smaller trees and older larger trees (Hadley and Veblen 1993). The outcome of this decrease in structural diversity of Douglas-fir is a temporary shift in dominance towards the seral pines. Thus, the overall rate of succession at the landscape scale is temporarily delayed until Douglas-fir populations can regain their prior status.

Relative to fire however, insect-induced increases in landscape heterogeneity are minor. Insect outbreaks are generally regional in extent (e.g., Swetnam and Lynch 1989; Baker and Veblen 1990) but selective in targeting specific tree species and tree sizes. By contrast, fires are often patchy and burn at varying intensities depending on local stand structures, fuel loadings, topography, and weather conditions (Arno 1980). Hence, the net result of fire at a large scale may be an increase in forest heterogeneity whereas at the stand scale fire may promote a more homogeneous forest structure.

Landscape scale interpretations of disturbance provide a unique perspective of processes constrained by stand structures yet predisposed by topographic variability and widespread anthropogenic forest alteration. In the Front Range, postfire stand development appears to be a significant factor determining the occurrence and severity of subsequent insect outbreaks within this landscape (Hadley and Veblen 1993). Hence, there is a great need to identify the topographic constraints of disturbance along with fire history into models and management policies that can deal with future effects of disturbance at the landscape scale.

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